

A Sigma Above Average: The Social Optimum in Mate Search Is Not an Equilibrium

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ABSTRACT

We study an agent-based model of mate search in which agents hold private, asymmetric preferences over one another and meet at random, forming couples and breaking them up as they go. Marriage enters as a commitment device: an agent stops searching once the affinity with its partner exceeds a threshold Λ . Imposed uniformly, the threshold has an interior optimum. A society that never commits keeps searching but keeps being abandoned; a society that commits immediately locks in its first acceptable partner; between the two, aggregate utility is maximised at $\Lambda^* = 1.001\sigma$, one standard deviation above the mean affinity. This yields what sounds like good advice: do not marry the first person you like and do not hold out for the love of your life, but marry if you like your partner more than a sigma above average.

Our main result is that no agent would take it. Holding the society at $\Lambda = \sigma$, a single agent's best threshold is 1.306 ± 0.005 , worth $+0.034 \pm 0.003$ against a typical utility of 1.45. Iterating the best response settles at a symmetric Nash equilibrium of $\Lambda_{eq} = 1.26$, where aggregate utility is 3.9% lower than at Λ^* . Commitment generates a positive externality — an agent that stops searching stops displacing other people's partners — which agents choosing alone do not internalise, so they systematically over-search. We show the gap is robust to the distribution of preferences and to whether being single is drawn from the same distribution as being coupled, and that it is unevenly distributed: once agents differ in how much they are liked, the least desirable gain most from the convention and the most desirable are made worse off by it. The advice is sound and the equilibrium does not deliver it.

KEYWORDS

Social norms; conventions; decentralised matching; search and stopping; commitment; price of anarchy; agent-based simulation

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1 INTRODUCTION

Consider the following situation: you are wondering whether to propose marriage to your partner. There are many aspects of marriage to consider, but you are mainly concerned with marriage as a *search strategy*. You like your partner, but you are not sure it is enough. There may be someone else out there whom you would prefer, and you are unlikely to meet that person if you marry. Perhaps

the best option is to stay open to the chance. But if your relationship is symmetric in this regard, you may find yourself single because your partner has moved on. While marriage removes some of the opportunities available to you, it also protects you from being left. So: is getting married a good search strategy?

Put that way, the question is about a *commitment threshold*. An agent searching for a partner in a market with no central clearing house faces a stopping problem, and marriage is a device for stopping: a level of affinity at which the agent binds itself to search no further. Call the level Λ . This paper studies what value Λ should take, and — the part that turns out to matter — whether agents left to choose it would take that value.

To make this precise we propose an agent-based model. A society of N agents holds private, asymmetric affinities for one another, drawn from a fixed distribution. At each of T steps, agents are paired at random; a pair becomes a couple when both strictly prefer the other to their current situation, and whoever is left behind returns to being single. An agent with a partner it likes by more than Λ is married, drops out of the matching process for good, and can neither leave nor be left.

As a uniform rule, Λ has an interior optimum. A society that never commits ($\Lambda = \infty$) keeps searching, but its agents keep being abandoned, and aggregate utility grows slowly. A society that commits at once ($\Lambda \ll 0$) locks in the first acceptable partner and does worse still. Between the two, aggregate utility is maximised at $\Lambda^* = 1.001\sigma$ in the Gaussian model — one standard deviation above the mean affinity. Hence the advice: *do not marry the first person you like and do not search for the love of your life, but get married if you like your partner more than a sigma above average*.

Our main result is that this is not something a society of self-interested agents would arrive at. Fix every agent at $\Lambda = \sigma$ and ask what a single agent should do. Its best threshold is 1.306 ± 0.005 , and playing it is worth $+0.034 \pm 0.003$ against a typical utility of 1.45 — a 2.3% gain, thirteen standard errors from zero. Moving everybody there and repeating, the iteration settles at a symmetric Nash equilibrium $\Lambda_{eq} = 1.26$, where aggregate utility is 3.9% below its value at Λ^* .

The mechanism is an externality. When an agent keeps searching and succeeds, it does not merely improve its own position: it displaces someone, who returns to being single and whose loss is not on the searching agent's books. Committing internalises that cost. An agent choosing its own threshold weighs only the option value of continued search against its own risk of being left, and so searches more than is collectively optimal. That the social optimum sits *below* the equilibrium — that everyone should be less picky than each would individually choose — is the content of the result.

Our contributions:

- We define a minimal model of decentralised mate search with a commitment threshold, and locate the welfare-optimal

threshold precisely: $\Lambda^* = 1.001\sigma$ for Gaussian affinities, 1.228σ for uniform ones (Section 5).

- We show this optimum is not an equilibrium. A single agent's best response to it is 1.306σ , and iterated best response converges to $\Lambda_{\text{eq}} = 1.26\sigma$ with a price of anarchy of 1.041 (Section 6).
- We show the welfare gain from the convention is unevenly distributed: once agents differ in desirability, the least liked gain most and the most liked are made worse off (Section 8).
- We show the optimum is robust to the shape of the affinity distribution and to decoupling the utility of being single from the utility of being coupled (Sections 5 and 7), and compare the decentralised outcome with Gale–Shapley and the utilitarian optimum (Section 9).

All results are reproducible; code and data accompany the paper.

2 RELATED WORK

The idea of using formal tools to study marriage is not new [6–8]. Following the taxonomy of models of the marriage market in [12], ours is a search model; see [11, 32, 34, 39] for search theory and [9, 10, 20, 29, 37, 38] for search models of marriage.

We differ from this work in two respects. Rather than analysing matching properties for generic preferences, we assume a distribution over preferences and specify a concrete meeting process. The motivation is the regime $N \gg T$, where the society is far larger than the number of people any agent meets. Compared with work that does not assume completely generic preferences [5, 26] we impose *less* structure in this way.

A famous relative is the stable matching problem [18], with a large literature on its structure and algorithmics [21, 23, 36] and on its use in market design [1, 33, 35]. In a society of millions, however, blocking pairs are guaranteed to exist and irrelevant, because the two agents will never meet. What matters is the process, not its fixed points; Section 3.2 notes that our system has no equilibrium configuration to converge to at all, which also limits the reach of dynamic notions of stability [2, 13–16, 19, 22, 24, 25, 27, 31]. A different route to the large-population limit of a matching problem is taken in [17].

The object we study — a uniform behavioural rule that is collectively good but individually resisted — is a social convention, and the question of whether a convention is self-enforcing is the standard one asked of conventions in multi-agent systems. Our model is therefore close in spirit to work in complexity economics [3, 4], and it illustrates a case where utility maximisation does not dictate observed behaviour: the dynamics follow simple behavioural rules, the resulting pattern is complex, and no agent has the computational power to work out which rule maximises aggregate utility.

3 THE MATE-SEARCH MODEL

The model contains N agents $\mathcal{A} = \{1, \dots, N\}$ with preferences over one another described by a utility function [28], encoded in the *affinity matrix* $A \in \mathbb{R}^{N \times N}$: A_{ij} is the utility for agent i of being in a couple with agent j . A is not symmetric in general, since i 's feelings for j need not match j 's for i , and A_{ij} is not positive in general, as it is common to meet people one would rather be single than date. We refer to the collection of agents as the *society*.

3.1 Affinities

In large societies the distribution generating A matters more than its values. Agent k 's preferences are the row $\{A_{kj} : j \neq k\}$; as with many human traits it is reasonable to take these as normally distributed. One might want different agents to have different parameters, but note that the k -th *column* $\{A_{jk} : j \neq k\}$ encodes how agent k is liked by society, and if rows were Gaussians with agent-specific parameters the columns would not in general be Gaussian. To keep rows and columns consistently Gaussian we take all off-diagonal entries from one distribution, $A_{kj} \sim \mathcal{N}(\mu, \sigma)$. We call this the *Gaussian model*, and for comparison also study the *uniform model*, $A_{ij} \sim U(\mu - 2\sigma, \mu + 2\sigma)$, which has the same standard deviation but no tails. Section 8 relaxes exchangeability and lets agents differ in how much they are liked.

The diagonal A_{kk} is the utility of being single. Its mean is the average utility of a society in which nobody has a partner, while the mean of the off-diagonal entries is the average utility if everyone were paired at random; the relationship between the two encodes how solitude compares with a random partner. We do not believe these are generically equal, but present our main results assuming they are and show in Section 7 that the conclusions survive when they are not.

3.2 Dynamics

The dynamics proceed in discrete steps $t \in \mathbb{N}$. At each step the agents still searching are paired at random — a uniformly drawn perfect matching π_t , with one agent sitting out if their number is odd. If agent k is single at t , so $u_t^k = A_{kk}$, and meets $l = \pi_{t+1}(k)$, the two form a couple if each prefers the other to its current situation:

$$u_{t+1}^k = \begin{cases} A_{kl}, & \text{if } A_{kl} > u_t^k \text{ and } A_{lk} > u_t^l, \\ u_t^k, & \text{otherwise.} \end{cases} \quad (1)$$

If instead k is in a couple with m , so $u_t^k = A_{km}$ and $u_t^m = A_{mk}$, and k meets $l = \pi_{t+1}(k)$ while m meets $n = \pi_{t+1}(m)$, then

$$u_{t+1}^k = \begin{cases} A_{kl}, & \text{if } A_{kl} > u_t^k \text{ and } A_{lk} > u_t^l, \\ A_{kk}, & \text{if } A_{mn} > u_t^m \text{ and } A_{nm} > u_t^n, \\ u_t^k, & \text{otherwise.} \end{cases} \quad (2)$$

The second line is the externality this paper is about: agent k loses its partner, and its utility drops all the way back to A_{kk} , because m found someone better. Nothing in m 's decision accounts for it. All comparisons are made against utilities as of the start of the step, and the system starts with every agent single, $u_0^k = A_{kk}$.

Running for T steps gives $\{u_t^k\}$, from which we take the average utility and its dispersion,

$$u_t \equiv \frac{1}{N} \sum_{k=1}^N u_t^k, \quad \sigma_t \equiv \left(\frac{1}{N} \sum_k (u_t^k)^2 - u_t^2 \right)^{1/2}. \quad (3)$$

Figure 1 shows the difference between the two levels of description. Individual trajectories jump up when an agent finds someone better and crash back to A_{kk} when an agent is left; the average climbs smoothly.

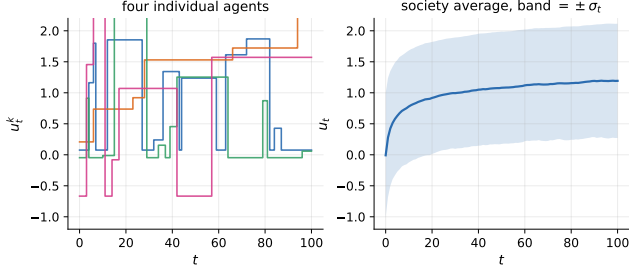


Figure 1: Left: utility of four agents, distinguished by colour. Right: average utility with a band of one standard deviation. Gaussian model $N(0, 1)$, $N = 10,000$, no commitment. The uniform model is visually indistinguishable.

3.3 Scale Invariance

Equations (1) and (2) depend only on inequalities between entries of A , so the dynamics are unchanged under $A_{ij} \mapsto \sigma A_{ij} + \mu$, and utilities transform the same way:

$$\text{if } (\mu = 0, \sigma = 1) \mapsto u_t^k \text{ then } (\mu, \sigma) \mapsto \sigma u_t^k + \mu. \quad (4)$$

We therefore report everything at $\mu = 0, \sigma = 1$; read 0 as μ and the vertical axes in units of σ . This is what makes “a sigma above average” a well-posed statement rather than a numerical coincidence.

3.4 No Configuration to Converge To

In large societies the number of people we meet in a lifetime is far smaller than the society, $T \ll N$, so the limit $T \rightarrow \infty$ is not the object of interest. It also does not exist in any useful sense. Consider three agents A, B, C where A most likes C , B most likes C , and C most likes A : at every step one is left out, and they remain in an eternal dance. The system never settles, which is why we study the trajectory of u_t at finite T rather than a steady state, and why notions of stability defined on configurations have limited purchase here. The equilibrium of Section 6 is an equilibrium in *thresholds*, not in matchings.

4 COMMITMENT

We add a *proposal vector* $\vec{\Lambda} \in \mathbb{R}^N$, where Λ_k is how much agent k must like a partner before committing. A couple is *married* once the affinity clears both thresholds:

$$(k, l) \text{ is married at } t \iff A_{kl} > \Lambda_k \text{ and } A_{lk} > \Lambda_l. \quad (5)$$

Writing $\mathcal{M}_t$ for the set of married agents, the matching at $t + 1$ excludes them:

$$\pi_{t+1} : \mathcal{A} \setminus \mathcal{M}_t \rightarrow \mathcal{A} \setminus \mathcal{M}_t. \quad (6)$$

A married agent is never matched again, so it can never be poached and never poaches: a married couple never breaks, $\mathcal{M}_t$ is non-decreasing, and $\Lambda = \infty$ recovers the model of Section 3 exactly.

Two readings of $\vec{\Lambda}$ run through the paper. In Sections 5 and 7 it is a *convention*: a single number Λ that every agent follows, characterising a society, and the question is which value is best for that society. In Section 6 it is a *strategy*: each agent chooses Λ_k for itself,

Table 1: Aggregate utility u_{100} by convention. Mean $\pm$ standard error over eight societies of $N = 10,000$.

Convention	Gaussian $N(0, 1)$	Uniform $U(-2, 2)$
$\Lambda = \infty$ (no commitment)	1.191 ± 0.004	1.269 ± 0.003
$\Lambda = 1.5$	1.298 ± 0.003	1.519 ± 0.001
$\Lambda = 1$	1.447 ± 0.003	1.543 ± 0.001
$\Lambda = 0.5$	1.285 ± 0.002	1.391 ± 0.001
$\Lambda = 0$	1.075 ± 0.003	1.255 ± 0.002
$\Lambda = -2$	0.868 ± 0.002	0.988 ± 0.002

and the question is which profile is stable. The gap between the answers is the paper’s result.

As in (4), results carry to generic μ, σ provided Λ is shifted and rescaled with the affinities, so $(\mu, \sigma, \sigma\Lambda + \mu) \mapsto \sigma u_t^k + \mu$. We measure Λ in units of σ above the mean throughout.

5 THE SOCIAL OPTIMUM

Unless stated otherwise, simulations use $N = 10,000$ agents, $T = 100$ steps, $\mu = 0$ and $\sigma = 1$, and quoted errors are standard errors over independent societies. Where a society with commitment is compared against one without, both run on the *same* affinity matrix with independent matchings, so the difference between them is not an artefact of sampling A .

5.1 An Interior Optimum

Figure 2 and Table 1 give aggregate utility as a function of the convention Λ . Two limits bracket the behaviour. For Λ above every entry of A , commitment never triggers and the society is that of Section 3: agents keep searching, and keep being left, and u_{100} reaches 1.191. For Λ below every entry, every agent marries the first agent it likes, and u_{100} falls to 0.868. Marrying the first person you like is worse than never marrying at all.

Between the extremes the optimum is interior. Resolving it on a grid of spacing 0.05 over 24 societies, and interpolating the peak through the top three points, gives

$$\Lambda^* = \begin{cases} 1.001 \sigma, & \text{Gaussian, } u_{100} = 1.4508 \pm 0.0013, \\ 1.228 \sigma, & \text{uniform, } u_{100} = 1.5784 \pm 0.0007. \end{cases} \quad (7)$$

For Gaussian affinities the optimum is one standard deviation above the mean to three digits. We do not have an argument for why it should be exactly σ , and the uniform model shows it is not a universal constant: with no tails the optimal threshold moves up by nearly a quarter of a sigma. What the two share is that the optimum is interior, and close enough to σ for the rule of thumb to be a good one in both.

Hence the advice, which we will spend Section 6 showing nobody would follow: *do not marry the first person you like and do not search for the love of your life, but get married if you like your partner more than a sigma above average.*

5.2 Why There Is an Optimum at All

The shape of Figure 2 comes from two opposing effects, and it is worth naming them because Section 6 turns on the fact that an individual agent sees only one of them.

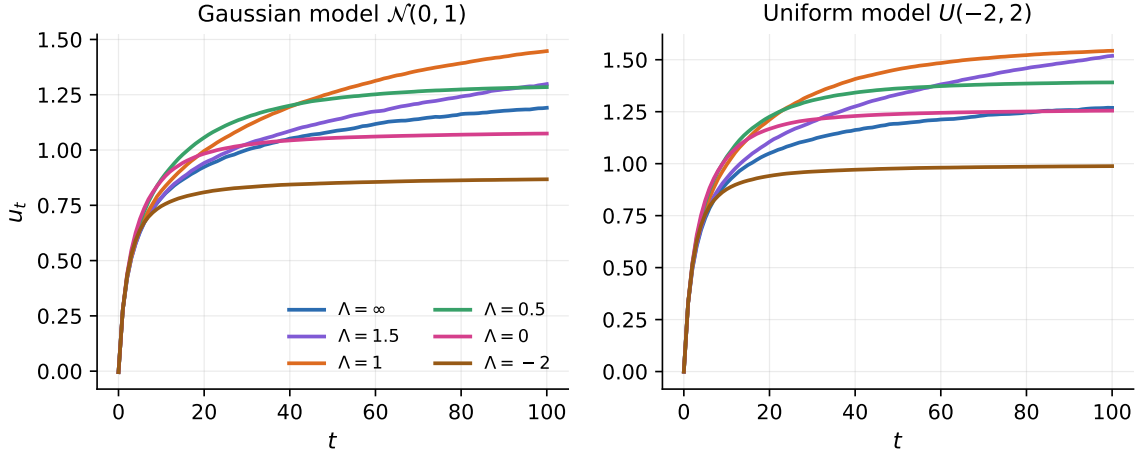


Figure 2: Aggregate utility in the Gaussian and uniform models with $N = 10,000$ for several values of the convention Λ ; $\Lambda = \infty$ is the society without commitment. Mean over eight independent societies with standard-error bands.

Raising Λ keeps an agent in the pool longer, which is good for that agent: more draws from p_A , and the best of many draws beats the best of few. This is the option value of search. But keeping an agent in the pool also keeps it available to poach, and every successful poach sends some third agent back to A_{kk} — the second line of (2). Aggregate utility is the sum over all agents, so it charges for the displacement. Lowering Λ removes agents from the pool, retiring both their option value and their capacity to displace others.

The optimum is where the two balance. An agent choosing for itself, however, weighs its own option value against its own risk of being left — and the displacement it inflicts on others appears nowhere in its payoff. That asymmetry is enough to predict the sign of the result that follows: agents should want a *higher* threshold than is socially optimal.

5.3 Commitment Also Reduces Inequality

Figure 3 plots u_t with its dispersion for the convention $\Lambda = \sigma$ against no commitment. The mean is higher and the variance *lower*: in the Gaussian model the standard deviation at $t = 100$ falls from 0.914 to 0.645, and in the uniform model from 0.838 to 0.371. If agents value equality of outcomes this is a second benefit on top of the first. The mechanism is visible in Figure 4: the convention imposes a floor, since every married agent is above Λ by construction, and it drains the mass of long-term singles sitting at their A_{kk} .

5.4 The Price: A Shorter Search

The gain is bought with something. Figure 5 counts the distinct partners each agent has over the run: without commitment an agent cycles through 5.15 partners on average in the Gaussian model, against 4.04 under the convention, and in the uniform model 5.00 against 2.76. An agent that does not commit ends up on average with someone it likes less, but may derive a surplus from having had more relationships — a surplus the model does not represent. Conversely, breaking up is generally painful, and that loss is not represented either, though including it would only add to the case for

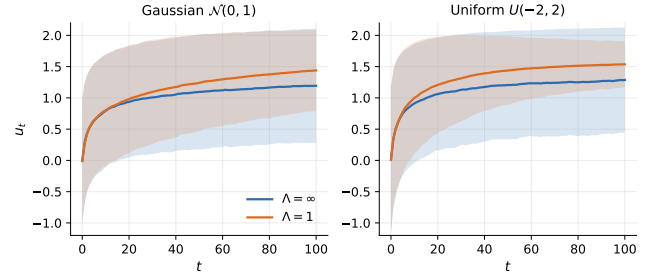


Figure 3: Aggregate utility with the convention $\Lambda = 1$ (orange) and without commitment (blue), with bands of one standard deviation, $N = 10,000$. Both societies share the same affinity matrix.

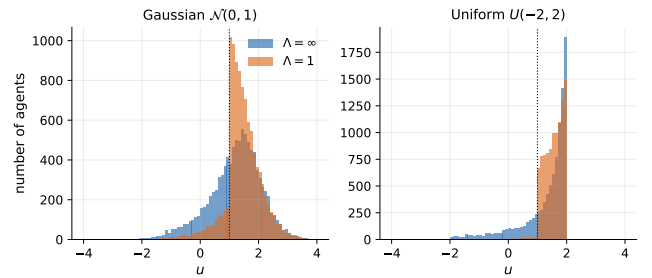


Figure 4: Histogram of $\{u_{100}^k\}$ with bin size 0.1 for the society with the convention $\Lambda = 1$ (orange) and without commitment (blue). Dotted line: the threshold.

commitment. Evidence suggests marriage is beneficial in a number of respects [40].

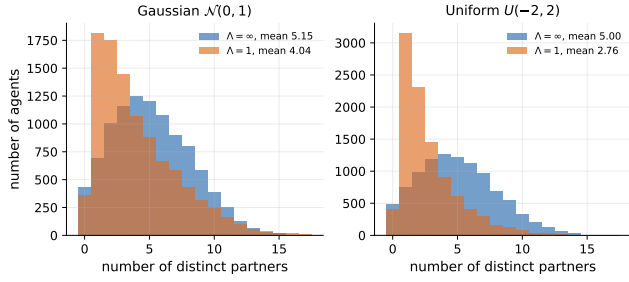


Figure 5: Distinct partners per agent over 100 steps, with and without the convention, $N = 10,000$.

6 THE EQUILIBRIUM THRESHOLD

Section 5 treats Λ as a convention handed to the society. Nothing so far makes an individual agent want to obey it. This section asks whether one would.

6.1 The Question and How We Measure It

Fix every agent to a common Λ_r and consider one agent choosing Λ' for itself. Because the society is exchangeable, “one agent” needs no qualification: there is no special agent to pick. Write $\Delta(\Lambda' | \Lambda_r)$ for the change in that agent’s utility at $t = T$ relative to conforming. The society is at a *symmetric Nash equilibrium* when

$$\Delta(\Lambda' | \Lambda_{eq}) \leq 0 \quad \text{for all } \Lambda', \quad (8)$$

that is, when no agent can do better by moving. Since N is large and one agent does not move the aggregate, this is a population game and (8) is its mean-field equilibrium condition. To locate it we use *iterated best response*: find the Λ' maximising $\Delta(\cdot | \Lambda_r)$, move the whole society there, and repeat, starting from $\Lambda_0 = \sigma$.

Measuring Δ requires care, because two pressures work against each other. A single agent is far too noisy: its utility at $t = T$ has a standard deviation of order σ , while the effect sought is a few hundredths of a utility. But many deviators are no longer one agent — let a large group deviate and it begins responding to itself rather than to the society, since a crowd of agents all holding out is a crowd of agents still in the matching pool, which changes the pool. We use 200 deviators per run, 2% of the society, scattered at random, with two safeguards.

Pairing. A deviator’s gain is measured against *the same agent* in a baseline run on the same affinity matrix with the same stream of matchings, so Δ is a mean of per-agent differences rather than a difference of two population means and the agent’s own luck in A cancels. Deviating to the convention must then come out identically zero, agent by agent, which is a test rather than a hope.

A witness for the environment. Every sweep also reports what the non-deviators got, differenced the same way. That number is the environment moving; if it is not zero, the measurement is not a best response to anything.

Replication is across societies, never across deviators: two deviators in one run share a matrix and a matching stream, so their gains are correlated. Each round below uses 512 independent societies, and every error bar is taken over them.

Table 2: Iterated best response from $\Lambda_0 = \sigma$, Gaussian model. “Gain” is the best unilateral improvement available at Λ_r ; the equilibrium condition (8) is that it be non-positive.

Round	Λ_r	Welfare u_{100}	Gain available
0	1.000	1.4487	+0.0337 $\pm$ 0.0025
1	1.306	1.3735	−0.0046 $\pm$ 0.0033
2	1.249	1.3953	−0.0006 $\pm$ 0.0011
3	1.259	1.3916	−0.0019 $\pm$ 0.0024
4	1.244	1.3972	−0.0003 $\pm$ 0.0023
5	1.253	1.3928	−0.0023 $\pm$ 0.0019
6	1.245	1.3973	+0.0048 $\pm$ 0.0021
7	1.254	1.3930	+0.0016 $\pm$ 0.0022

6.2 Result

Round 0 answers the question on its own. With the whole society at $\Lambda = \sigma$, a single agent’s best threshold is 1.306 ± 0.005 , not 1.000, and playing it is worth $+0.0337 \pm 0.0025$ against a typical utility of 1.45 — a 2.3% gain, thirteen standard errors from zero. A lone agent is better off holding out for a partner it likes rather more than the convention asks. The sign is the one predicted in Section 5: agents want to over-search, because the displacement they cause is not on their books.

The iteration settles at

$$\Lambda_{eq} = 1.259 \pm 0.022, \quad (9)$$

the spread being over the settled rounds. It arrives in a single step — the jump at round 0 — and everything after is jitter at the measurement noise of the argmax, because the best response is nearly flat in the convention it answers. The column that establishes convergence is the last one, not the trajectory: a fixed point is reached when no agent can do measurably better by moving, and from round 1 onward the available gain is statistically zero. The iterate sitting still is the weaker test; the incentive vanishing is the real one.

6.3 What It Costs

Aggregate utility peaks at round 0, exactly where Section 5 says it should, and falls once agents are allowed to choose:

$$\begin{aligned} u_{100}(\Lambda^*) &= 1.4487 \pm 0.0003, \\ u_{100}(\Lambda_{eq}) &= 1.3915, \\ \text{price of anarchy} &= 1.041. \end{aligned} \quad (10)$$

Letting agents pick their own thresholds costs 0.057 of utility, or 3.9% of the aggregate. Every agent is worse off at Λ_{eq} than at Λ^* — this is a symmetric equilibrium of a symmetric game, so there is no redistribution hiding inside the average and no agent that gains from the move.

The structure is that of a commons. Each agent searching a little longer is individually right and collectively wrong, and the society lands at a threshold 26% above the one that would serve it best. The advice of Section 5 is sound as advice; what the model says is that it will not be arrived at by agents optimising alone, which is the usual reason societies enforce conventions rather than leaving them to individual judgement.

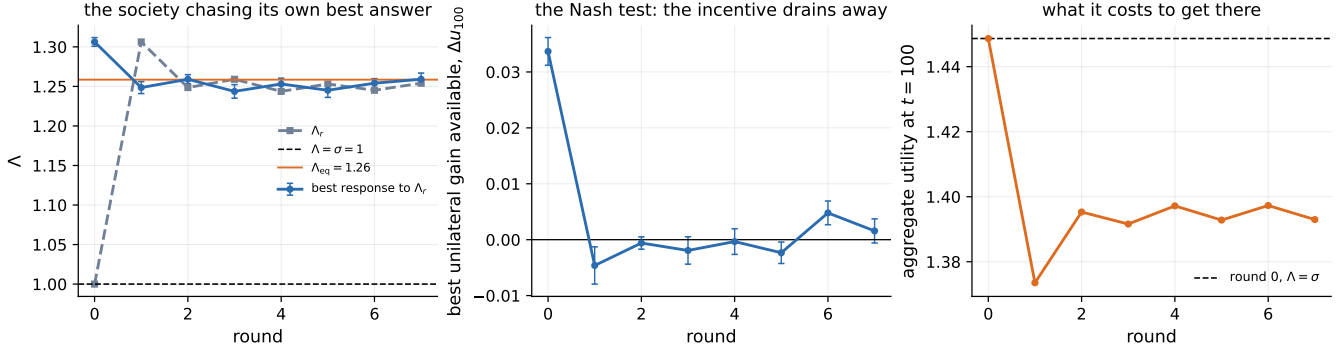


Figure 6: Iterated best response from the convention $\Lambda = \sigma$, Gaussian model, $N = 10,000$, 512 societies per round. Left: the society chasing its own best answer. Centre: the best unilateral gain available, which is the Nash test — it is thirteen standard errors from zero at round 0 and statistically zero from round 1 on. Right: aggregate utility, which peaks at round 0 and can only fall thereafter.

6.4 Reading the Equilibrium

Two remarks on what Λ_{eq} is and is not.

It is an equilibrium in *thresholds*, not in matchings. The underlying matching process never settles (Section 3.2); what settles is the rule agents use to stop. This is why the equilibrium concept here is the appropriate one even though stable matchings are not.

It is also the equilibrium of the game at horizon $T = 100$. Both Λ^* and Λ_{eq} depend on the horizon, since a society expecting a long search should be more demanding than one expecting a short one — visible in Figure 2 as the curves crossing. We fix $T = 100$ throughout; the horizon-dependence of the gap is left for future work.

7 ROBUSTNESS

7.1 When Being Single Is a Different Draw

Section 3.1 assumed for concreteness that the diagonal and off-diagonal entries of A come from the same distribution. To check the conclusion does not depend on it, we set $A_{kk} = 0$ for every agent — being single is worth exactly nothing — and draw $A_{ij} \sim \mathcal{N}(\mu, 1)$ for $i \neq j$, sweeping μ with the convention tracking the distribution at $\Lambda = \mu + \sigma$. Negative μ means most people are worse than solitude; positive μ means almost anyone beats it.

Writing $f(\mu) = u_M(\mu) - u(\mu)$ for the gap between the society with the convention and the society without, Figure 7 gives $f(-2) = +0.001$, $f(-1) = +0.027$, $f(0) = +0.283$, $f(1) = +0.565$, $f(2) = +0.739$: positive throughout and non-decreasing in μ . The ends of the sweep also cover what lies outside it. Below $\mu = -2\sigma$ an agent rarely meets anyone it prefers to solitude and the system behaves as at $\mu = -2$; above $\mu = 2\sigma$ agents prefer almost any partner to solitude, as at $\mu = 2$, up to the change of scale set by μ . The benefit of commitment is therefore not an artefact of the two distributions coinciding.

7.2 Men, Women, and Other Partitions

We did not distinguish men from women, as in the original marriage problem [18], which might suggest we are dealing with stable roommates instead. The distinction does not matter here. Partitioning the society in two and matching only across the partition

gives curves indistinguishable from Figure 3: over four societies the two-sided variant differs from the one-sided one by less than 0.03, inside the run-to-run spread.

Partitions of unequal size work too, with the matching becoming a random injection from the smaller side into the larger and a random subset of the larger sitting out each step. Further variants are available — allowing a subset of one partition to match within itself as well admits a reading as bisexual agents. This flexibility contrasts with the matching problem, where many results including Gale–Shapley require the two sides to be the same size [36].

8 WHO GAINS AND WHO LOSES

The equilibrium of Section 6 is symmetric because the society is exchangeable: every agent faces the same distribution, so every agent has the same best response. Real societies are not like that, and relaxing exchangeability changes who the convention is for.

It is tempting to look for the effect inside the model as it stands, by ranking agents on the mean of their column of A — how much society likes them — and splitting off the top and bottom percentiles. This asks more of finite-size effects than they can deliver. With $N = 10,000$ i.i.d. entries, column means are spread by only $1/\sqrt{N} \approx 0.01\sigma$, so the “1% most liked” are more liked by about 0.03σ , against a commitment benefit of 0.25σ . Over six societies the premium is $+0.219 \pm 0.045$ for the least liked and $+0.254 \pm 0.043$ for the most liked: indistinguishable. Percentiles exist there as luck of the draw, not as a property of an agent.

To ask the question properly the model needs genuine *vertical* heterogeneity. We add a per-agent desirability to every entry of that agent’s column,

$$A_{ij} = \varepsilon_{ij} + q_j, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon), \quad q_j \sim \mathcal{N}(0, \sigma_q), \quad (11)$$

holding $\sigma_\varepsilon^2 + \sigma_q^2 = 1$ so that runs at different σ_q stay comparable. Now some agents really are more widely liked, and σ_q tunes how much.

Figure 8 and Table 3 give the result. The convention helps the less desirable more, and past $\sigma_q \approx 0.4$ its effect on the most desirable turns *negative*: they are made strictly worse off by living in a society

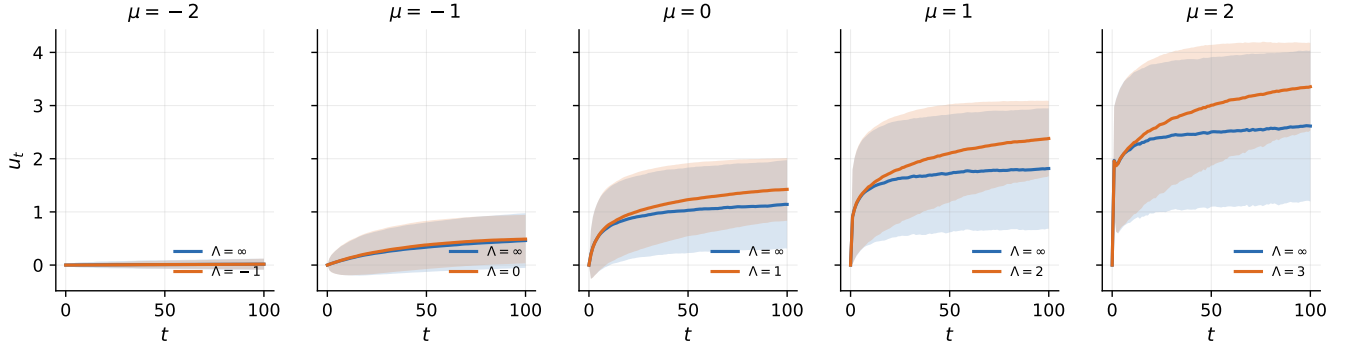


Figure 7: Aggregate utility with $A_{kk} = 0$ and $A_{ij} \sim \mathcal{N}(\mu, 1)$ for $i \neq j$, $N = 10,000$. Orange: the convention $\Lambda = \mu + 1$. Blue: no commitment.

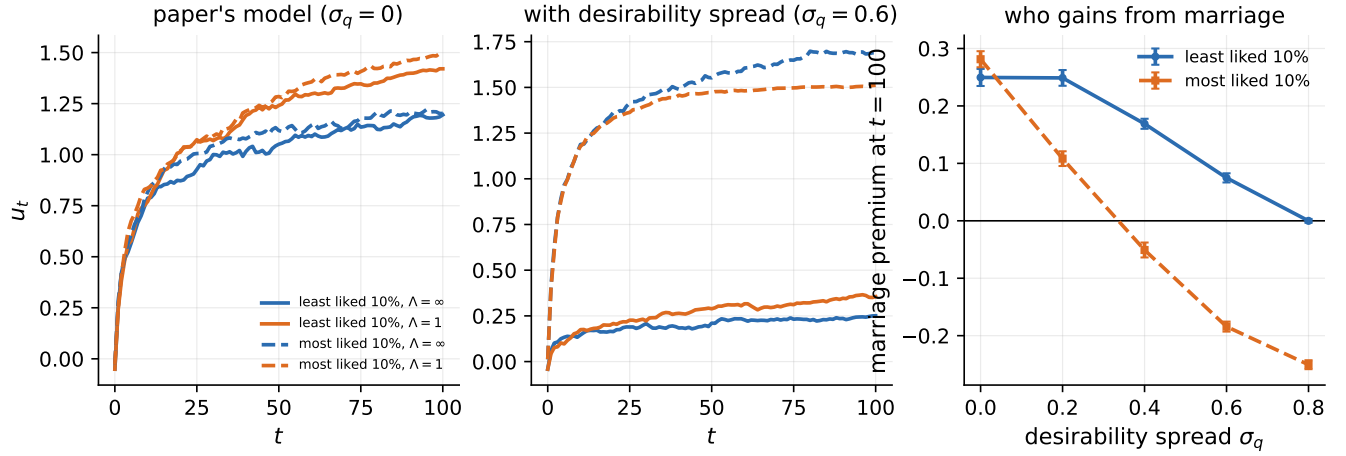


Figure 8: Benefit of the convention $\Lambda = \sigma$ at $t = 100$ for the most and least liked 10% of agents. Left: the exchangeable model, $\sigma_q = 0$, where the two are indistinguishable. Centre: with desirability spread $\sigma_q = 0.6$. Right: the benefit against σ_q , mean $\pm$ standard error over six societies.

Table 3: Benefit of the convention, $u_M - u$ at $t = 100$, by desirability spread, for the most and least liked 10%. Mean $\pm$ standard error over six societies.

σ_q	Least liked 10%	Most liked 10%
0.0	$+0.250 \pm 0.015$	$+0.281 \pm 0.014$
0.2	$+0.249 \pm 0.014$	$+0.108 \pm 0.013$
0.4	$+0.169 \pm 0.009$	-0.051 ± 0.013
0.6	$+0.075 \pm 0.008$	-0.184 ± 0.009
0.8	-0.000 ± 0.003	-0.251 ± 0.008

that commits. The reading follows from Section 5. Commitment is insurance against being left. Agents whom everyone likes are rarely at risk of being left, so the insurance is worth little to them, while the option value they give up — the chance of meeting someone better, which for them is a realistic prospect — is worth a great deal.

This compounds the problem of Section 6 rather than relieving it. There, a uniform society over-searches relative to its own optimum. Here, the agents with the strongest private reason to defect from the convention are precisely the ones it was not helping anyway, so enforcing it faces resistance from exactly the quarter with the least to lose by resisting.

9 CENTRALISED BENCHMARKS

How far is any of this from what a planner could do on the same affinity matrix? We use three reference points, computed by splitting the society into equal halves and matching across them, which is the setting Gale–Shapley is defined for: random pairing, the proposer-optimal stable matching [18] with average utility u_{G-S} , and the matching maximising total surplus [6, 7],

$$u(\pi) = \frac{1}{2} \sum_{i=1}^N (A_{i\pi(i)} + A_{\pi(i)i}). \quad (12)$$

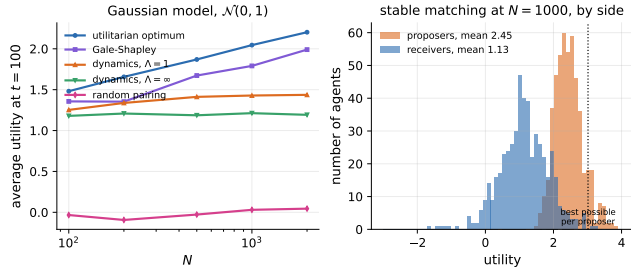


Figure 9: Left: aggregate utility at $t = 100$ for the dynamics against random pairing, the stable matching and the utilitarian optimum, as a function of N . Right: the two sides of the stable matching at $N = 1000$ do very different things.

Naively (12) ranges over $N!$ matchings, which for $N \approx 1000$ has 2568 digits. But it is a linear assignment problem on the symmetrised surplus $A_{ij} + A_{ji}$, so the Hungarian algorithm solves it exactly in $O(N^3)$; we report it up to $N = 2000$.

At $N = 1000$ in the Gaussian model the dynamics reach 1.213 without commitment and 1.429 under the convention, the stable matching reaches $u_{G-S} = 1.791$, and the utilitarian optimum 2.045. Random pairing gives 0.030, that is μ , as it must. The planner’s margin grows with N , since both centralised benchmarks have more to work with as the society grows while the dynamics are limited by T .

The gap is a second, larger sense in which the decentralised market is sub-optimal, on top of the 4.2% of Section 6. If affinities could be inferred from manifest, correlated properties — age, personality, political leaning, education — a central mechanism could propose a stable matching and do considerably better than any convention.

The right panel adds a caveat about the size of that margin. A proposer-optimal stable matching is far from symmetric between sides: at $N = 1000$ the proposing side averages 2.454 and the receiving side 1.128, and u_{G-S} is the average of the two. The mean best-possible partner for a proposer is 3.015, so no matching of any kind places the two-sided average near it. Claims about the planner’s advantage should say which side of the market they mean.

10 EMPIRICAL COMPARISON

Aggregate utility is hard to measure in a real society; the number of married people is not. Figure 10 places the model’s married share at each step beside the share of men in England and Wales ever married by a given age, by birth cohort [30].

The agreement is reasonable given how much the model ignores — divorce, most obviously — and it was not built in: the resemblance is a dynamical consequence of the matching process. Read through the model, the declining married share across cohorts is a society-wide rise in Λ , which agrees with rough expectations: marriage was once enforced by tradition and many couples married without high affinity, and as those forces weakened people became more demanding.

Section 6 suggests a reading of that drift which the data alone does not supply. If a society’s Λ is maintained by convention, then

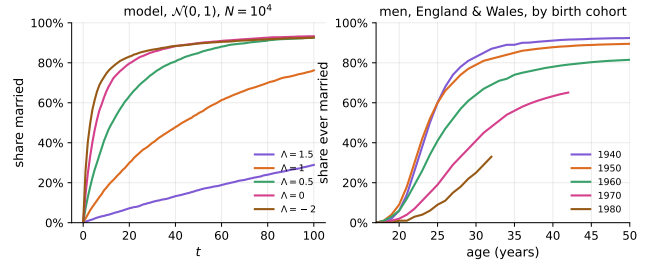


Figure 10: Left: share of married agents in the Gaussian model with $N = 10,000$, for $\Lambda = 1.5, 1, 0.5, 0, -2$ from the bottom line upwards. Right: share of men in England and Wales ever married by a given age, for the birth cohorts 1980, 1970, 1960, 1950, 1940 from the bottom line upwards [30].

weakening the convention should move Λ towards the value individual agents prefer — which is *higher* than the social optimum, by 26% in our simulations. A secular rise in Λ is what the model predicts a society will do when it stops enforcing a threshold and lets agents choose, and the prediction is that aggregate utility falls as it happens. We offer this as an interpretation, not a finding: the model has no account of why conventions weaken, and the comparison below is visual rather than fitted.

Two caveats bound how far Figure 10 can be pushed. Steps are not years: the model’s curve is concave from $t = 0$ while the data is S-shaped, because nobody marries before their late teens, and any quantitative cohort-to- Λ mapping needs an explicit step-to-age map, which we do not have. And the association of particular cohorts with particular Λ values is an eyeball match, not a fit.

11 DISCUSSION AND LIMITATIONS

The benefits of commitment disclosed here concern a narrow aspect of marriage, and may be irrelevant once others are considered; Section 8 shows that even within the model the benefit is not uniform across agents.

The most salient simplification is that affinities are time-independent. Making them time-dependent would bring divorce in for free: a married couple’s affinity may fall below Λ , at which point they unmarry and may break apart if one finds another mate. It would also account for spontaneous break-ups, which require only that affinity fall below one partner’s utility of being single. This is the extension most likely to connect the model to further data.

Three further limitations. The meeting process is uniform over the whole society, whereas real meeting is strongly assortative by geography, education and social network. Both Λ^* and Λ_{eq} are computed at the fixed horizon $T = 100$, and we have not characterised how the gap between them varies with T . And the equilibrium of Section 6 is located numerically, by iterated best response over a grid; we have no closed-form characterisation of Λ_{eq} , nor a proof that it is unique.

12 CONCLUSIONS

We studied an agent-based model of decentralised mate search in which marriage is a commitment threshold Λ . As a uniform convention, Λ has an interior optimum: never committing and committing immediately are both worse than committing at $\Lambda^* = 1.001\sigma$, one standard deviation above the mean affinity. Commitment raises aggregate utility and lowers its dispersion at the same time. This is the source of what sounds like good advice — do not marry the first person you like and do not search for the love of your life, but marry if you like your partner more than a sigma above average.

The advice is not self-enforcing. Holding the society at $\Lambda = \sigma$, a single agent does better by holding out for 1.306σ , and iterated best response settles at $\Lambda_{eq} = 1.26\sigma$, where everyone is 3.9% worse off. The cause is an externality: an agent that keeps searching and succeeds displaces somebody, and that cost appears in the aggregate but not in its own payoff. Agents optimising alone therefore over-search, and the society lands above the threshold that would serve it best. Once agents differ in how much they are liked, the convention's benefit concentrates on the least desirable and turns negative for the most desirable, so the pressure against it comes from the quarter with the least to lose.

The model leaves several directions open. A closed-form characterisation of Λ_{eq} , and a proof of uniqueness, would put Section 6 on firmer ground. The dependence of both Λ^* and Λ_{eq} on the horizon T is unexplored and is where the model makes contact with the observation that different societies sustain different thresholds. Letting agents *learn* Λ from their own history, rather than best-responding with full knowledge, would test whether the equilibrium is reachable by agents with realistic information — Section 10 sketches the rule such an agent might use. And a time-dependent affinity would bring divorce, and with it divorce rates, within reach of the same framework.

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